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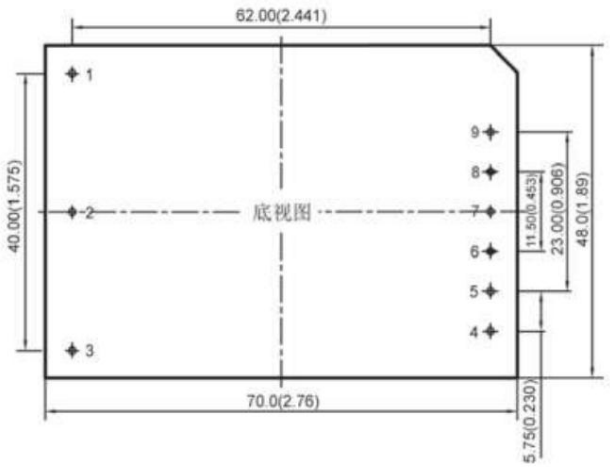
TDPOWER

TLH50_R2 Series AC-DC Power Supply



Features

ÿWide input voltage: 85ÿ265VAC, 100ÿ370VDC, 47ÿ400Hz

ÿProducts with an ultra-wide voltage input range can be customized with an "E" suffix.

ÿHigh reliability, high-precision regulated output, low ripple noise

ÿRated power 50W, plastic or metal casing

ÿInternational standard package size: DIP 70*48*23mm, compatible with multiple manufacturers.

ÿ No-load power consumption as low as ÿ0.15W, full-load efficiency range of 84%ÿ88%

ÿ Meets single/dual channel requirements, isolation withstand voltage ÿ2500VAC

ÿInput undervoltage protection, output overcurrent protection, output short circuit protection with self-recovery.

ÿOperating temperature: -40ÿÿ+85ÿ

ÿ CE certified; lead-free standards must be specified when placing an order.

ÿSupport 100% domestic products

ÿThree-year warranty

Application Scope

ÿThe TLH50_R2 series power supply is an AC/DC dual-use power module with the same package size as the TA50 series. It supports 100% domestic production and comes with a three-year warranty; input range...

85ÿ265VAC/100ÿ370VDC. For products requiring an ultra-wide input range of 85V-310VAC/100V-435VDC, customized production with the suffix "E" is available upon request.

Product specifications; Full-load output power 50W (plastic casing or with metal casing); No-load power consumption as low as 0.15W, efficiency range 84%ÿ88%; Meets single/dual output requirements, isolation withstands...

Voltage ÿ2500VAC; Operating temperature -40ÿÿ+85ÿ; Input undervoltage protection, overcurrent protection, output short circuit protection with self-recovery.

ÿThis series is widely used in industries such as power, new energy, instrumentation, Internet of Things, industrial control, office and civil use.

ÿWhen the application has high electromagnetic compatibility requirements, the EMC peripheral application circuit of this series of products must be referred to.

Input characteristics

project	illustrate
Input voltage range	85ÿ265VAC, can withstand 280VAC voltage surge for short periods.
Input frequency range	47ÿ400Hz
Recommended Fuse	3A slow-blow fuse
standby power consumption	ÿ0.15W

Output characteristics

project	illustrate
Output voltage accuracy	Vo1ÿ±2%
	Vo2ÿ±5%
Linear adjustment rate	ÿ0.5%
Load regulation	ÿ1%
Temperature coefficient	ÿ0.02%/ÿ
Short circuit protection	Long-term short circuit protection, self-recovery
Overcurrent protection	ÿ120%

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project	illustrate
Isolation voltage	ÿ2500VAC
Insulation resistance	500VDC,ÿ100Mÿ
Switching frequency	Typical 65kHz
Operating temperature range	-40ÿ85ÿ
Storage temperature range	-40ÿ105ÿ
Storage humidity	ÿ95%RH
welding temperature	Manual welding: 350-400ÿ, time ÿ5 seconds
	Wave soldering at 260±5ÿ for 5–10 seconds
Cooling method	Natural cooling
Mean Time Between Failures (MTBF)	200000h
isolation capacitor	1000pF
Shell material	plastic shell

EMC characteristics

I	Transmitted harassment	EN55011(CISPR11) / EN55032(CISPR32ÿCLASS B
	Radiated disturbance	EN55011(CISPR11) / EN55032(CISPR32ÿCLASS B
EMS	electrostatic discharge	IEC/EN61000-4-2
	Radiated immunity	IEC/EN61000-4-3
	Impulse group immunity	IEC/EN61000-4-4
	Surge immunity	IEC/EN61000-4-5
	Conducted disturbance immunity	IEC/EN61000-4-6
	Voltage dips, drops and short interruptions immunity	IEC/EN61000-4-11

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Selection List

model	Input voltage VAC	Output voltage Vo1	Output Vo2	Output current Io1	Output Io2	Ripple noise ÿmVÿ	Typical efficiency	Maximum capacity uF
TLH50-23B05R2	85ÿ265VAC	5.05		10		50	84%	4700
TLH50-23B12R2	85ÿ265VAC	12		4.17		100	88%	3300
TLH50-23B15R2	85ÿ265VAC	15		3.34		100	87%	2200
TLH50-23B24R2	85ÿ265VAC	24		2.1		100	88%	1000

Note 1: TLH50-23B05R2, “B” represents a single-channel output voltage of 5V.

Note 2: This series has an internal primary filter circuit. For higher levels of lightning surge testing or Class IV EMC testing, the parameters of external components need to be adjusted. Please contact sales.

The department obtains peripheral recommended circuits.

Note 3: For products requiring an ultra-wide input voltage range of 85V-310VAC/100V-435VDC, products with the suffix "E" can be customized and manufactured according to requirements. Please contact the sales department.

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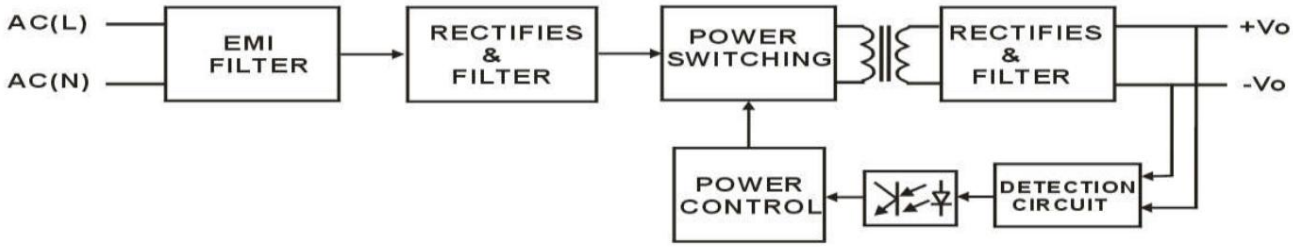
schematic diagram



Typical application diagram

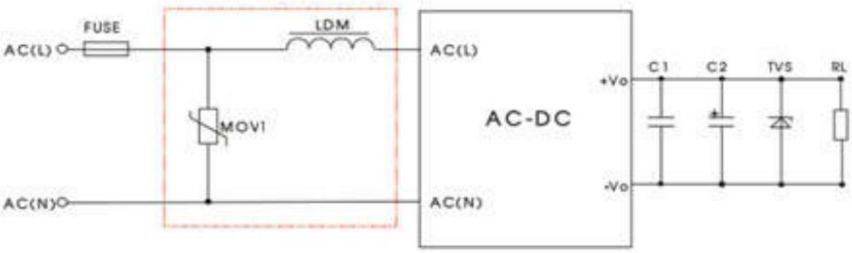


Figure 1 General Application Circuit

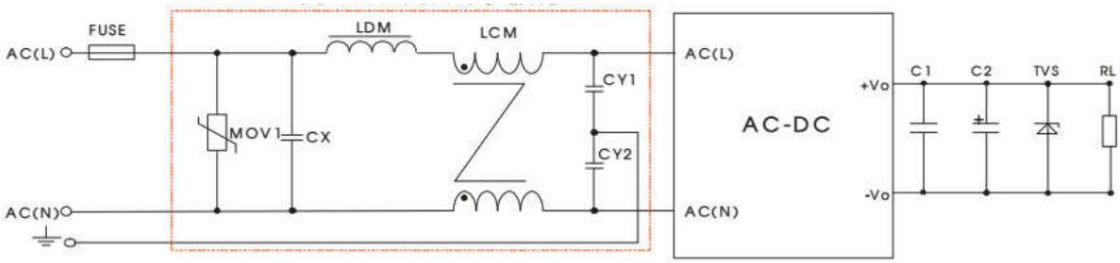


Figure 2 EMC Solutions

Typical values of external circuit components

Product Model\ Component	FUSE	MOV1	CX	LDM	LCM	CY1,CY2	C1	C2	TVS
TLH50-23B05R2	T2A/250V	MOV is a piezoelectric Resistance, recommended value 14D471K/ 20D471K Electricity	CX is a safety-compliant electric vehicle. Allow, 104K/275V	LDM is a differential mode Inductor, recommended The value is 47UH	LCM for common Modular inductor, push Recommendation value 10Mh~30Mh Recommended use UF9.8-5mH	Y1:102M/ 400VAC	105K/50V (ceramic tile) Allow)	470uF/16V	SMBJ7.0A
TLH50-23B12R2								220uF/25V	SMBJ15A
TLH50-23B15R2								220uF/25V	SMBJ18A
TLH50-23B24R2								100uF/35V	SMBJ27A

Note: If you need to pass the Level 4 EMC test, the parameters of the external components need to be adjusted. Please contact the sales department for specific parameters.

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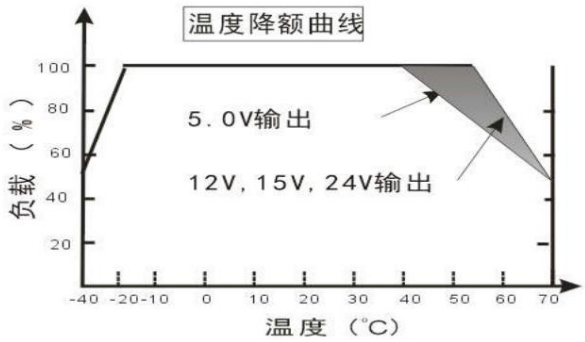
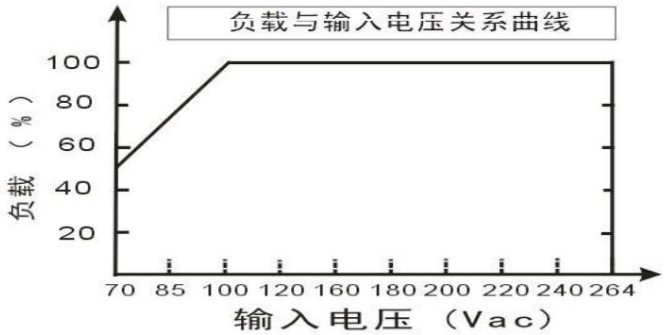
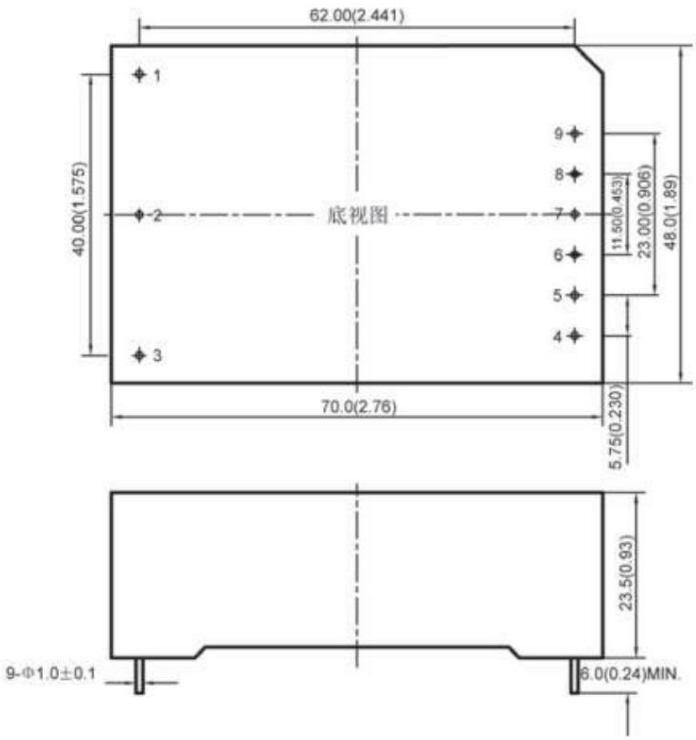




Structure diagram and pin definitions



Model/Pin Number	1	2	3	4	5	6	7	8	9
TLH50-23BxxR2 Single Output	L	N	FG	E.G	In-	E.G	E.G	E.G	In+
TLH50-23AxxR2 Dual Output	L	N	FG	E.G	Vo2-	E.G	WITH	E.G	Vo1+

Note: NP is a pinless pin, NC is an empty pin, FG is a pin with no internal connection, L/N pins have no wiring order requirement, and COM is the common ground for dual outputs.

Packaging information: 12 pieces per box, 15 boxes per carton, totaling 180 pieces.

Weight information: Approximately 105g/piece, gross weight of one box is approximately 1.3Kg, gross weight of one carton is approximately 20Kg.

Please specify if you require lead-free standards when placing an order.